

CASE STUDY

Real Estate Pricing & Geographic Micro-Market Clustering

Name	Gurugubelli Manoj Bharadwaj
Roll No	A24126552206
IBM ID	IBMQ2DST1386
Department	CSE (AI & ML)
College Name	Anil Neerukonda Institute of Technology and Sciences
Module Name	AI and ML
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Case Study Opted

Case Study Title: Real Estate Pricing & Geographic Micro-Market Clustering

House prices depend heavily on local geographic micro-markets that go beyond simple neighborhood or administrative boundaries such as ZIP codes or counties. Two houses that are structurally identical can differ significantly in value simply because they sit in different micro-markets — one near a coastline with high demand, and another inland with different economic drivers. This case study designs and builds a system that clusters properties by location and structural similarity into meaningful micro-markets, and then fits specialized pricing regression models within each cluster to predict real estate values more accurately than a single, one-size-fits-all model.

The project uses the California Housing Prices dataset, a well-known real-world dataset derived from the 1990 U.S. Census, containing block-group-level housing and demographic data across the state of California. The approach combines unsupervised learning (clustering) with supervised learning (regression) in a two-stage pipeline: first segment the state into geographically and structurally coherent micro-markets, then train a dedicated regression model per micro-market to capture the local pricing dynamics that a single global model would average away.

1. Overview

A single global regression model trained on statewide data tends to underperform because it forces one set of coefficients or one tree ensemble to explain price behavior across radically different housing markets — dense urban coastal cities, quiet inland suburbs, agricultural central-valley towns, and desert or mountain communities. Each of these micro-markets has a different relationship between price and features such as income, proximity to the ocean, population density, and housing age. Averaging across all of them blurs these local signals and increases prediction error, particularly at the high and low ends of the price range.

The goal of this case study is to first segment California properties into geographically and structurally coherent micro-market clusters using unsupervised clustering, and then train specialized regression models within each cluster so that local pricing patterns are captured explicitly rather than averaged away by a single statewide model.

- Tools Used: Python (pandas, NumPy, scikit-learn, XGBoost, GeoPandas, Matplotlib, Seaborn)
- Dataset: California Housing Prices Dataset (derived from the 1990 U.S. Census, sourced via the StatLib repository / Kaggle)
- Deliverable: A clustering pipeline that segments properties into micro-markets, a set of per-cluster regression models, and a comparison report benchmarking localized models against a single global baseline model

2. Business Problem

Real estate valuation and investment decisions — whether made by appraisers, mortgage lenders, investors, or online estimator tools — typically rely on regression models trained across broad geographies. This creates three recurring problems:

- Overgeneralized pricing models: A single statewide (or even city-wide) model assumes that the relationship between features like income, room count, or age and final price is constant everywhere, when in reality it varies sharply by micro-market.

- Fixed administrative boundaries don't reflect true micro-markets: ZIP codes, counties, and census tracts are drawn for administrative convenience, not for economic or housing-market homogeneity. Two adjacent ZIP codes can belong to very different real housing markets, while two non-adjacent regions can behave similarly.
- Structural feature importance varies by region: Ocean proximity may be the dominant price driver in a coastal cluster, while median income or housing age may dominate in an inland or rural cluster. A single global feature-weighting scheme cannot capture this.

Business Goal: Design a clustering-plus-regression pipeline that segments California into data-driven micro-markets and measurably improves prediction accuracy (target: reduce RMSE by at least 15–20% versus a single global regression baseline) so that pricing tools, appraisal systems, and investment models can rely on locally accurate, defensible estimates.

3. Dataset

The California Housing Prices dataset contains 20,640 records, each representing a census block group in California, with features covering geography, housing characteristics, and demographics. The dataset was originally compiled from the 1990 U.S. Census and is widely used as a benchmark regression dataset because it combines real geographic coordinates with meaningful structural and socioeconomic features.

Sample Data (First 6 Rows)

longitude	latitude	housing_median_age	total_rooms	median_income	median_house_value	ocean_proximity
-122.23	37.88	41	880	8.3252	452,600	NEAR BAY
-122.22	37.86	21	7,099	8.3014	358,500	NEAR BAY
-118.30	34.05	35	1,467	3.5417	215,300	<1H OCEAN
-119.72	36.75	18	2,015	2.0956	89,400	INLAND
-117.15	32.72	29	2,491	4.6512	312,700	NEAR OCEAN
-121.50	38.62	16	3,340	3.8125	168,900	INLAND

Table 1: Sample rows from the California Housing Prices dataset

Data Dictionary

Column Name	Description	Data Type
longitude	Geographic longitude of the block group	Float
latitude	Geographic latitude of the block group	Float
housing_median_age	Median age of houses within the block group (years)	Integer
total_rooms	Total number of rooms across all households in the block group	Integer
total_bedrooms	Total number of bedrooms across all households (contains missing values)	Integer

Column Name	Description	Data Type
population	Total population residing within the block group	Integer
households	Total number of households (occupied housing units) in the block group	Integer
median_income	Median household income of the block group (in tens of thousands USD)	Float
ocean_proximity	Categorical distance-to-ocean descriptor (e.g., NEAR BAY, INLAND, ISLAND)	Categorical
median_house_value	Target variable — median house value for the block group (USD)	Float

Table 2: Column-level data dictionary for the dataset

4. Feature Engineering & Geographic Clustering Approach

4.1 Data Cleaning & Preprocessing

- Missing values: total_bedrooms contains approximately 207 missing entries; these were imputed using the median value within each preliminary geographic grid cell rather than a single global median, to avoid distorting local density ratios.
- Outlier capping: median_house_value is capped at 500,001 in the raw data (a census reporting artifact); these capped records were flagged and evaluated separately to avoid skewing model training.
- Categorical encoding: ocean_proximity was one-hot encoded for use as a structural feature alongside clustering.
- Feature scaling: StandardScaler was applied to all continuous features prior to clustering, since K-Means is distance-based and sensitive to differing feature scales.

4.2 Derived Features

- $\text{rooms_per_household} = \text{total_rooms} / \text{households}$
- $\text{bedrooms_per_room} = \text{total_bedrooms} / \text{total_rooms}$
- $\text{population_per_household} = \text{population} / \text{households}$

These ratio features capture housing density and household composition more effectively than the raw counts, and were found to correlate more strongly with price than their unscaled counterparts.

4.3 Clustering Method

K-Means clustering was applied to a feature set combining geographic coordinates (latitude, longitude) with density and income features (population_per_household, median_income), so that clusters reflect both physical location and underlying socioeconomic character — not geography alone. This produces clusters that behave as genuine micro-markets rather than arbitrary geographic grids. DBSCAN was evaluated as an alternative for detecting irregularly shaped, density-based clusters, but K-Means was selected as the primary method for its interpretability and stability across runs, with DBSCAN retained as a robustness check.

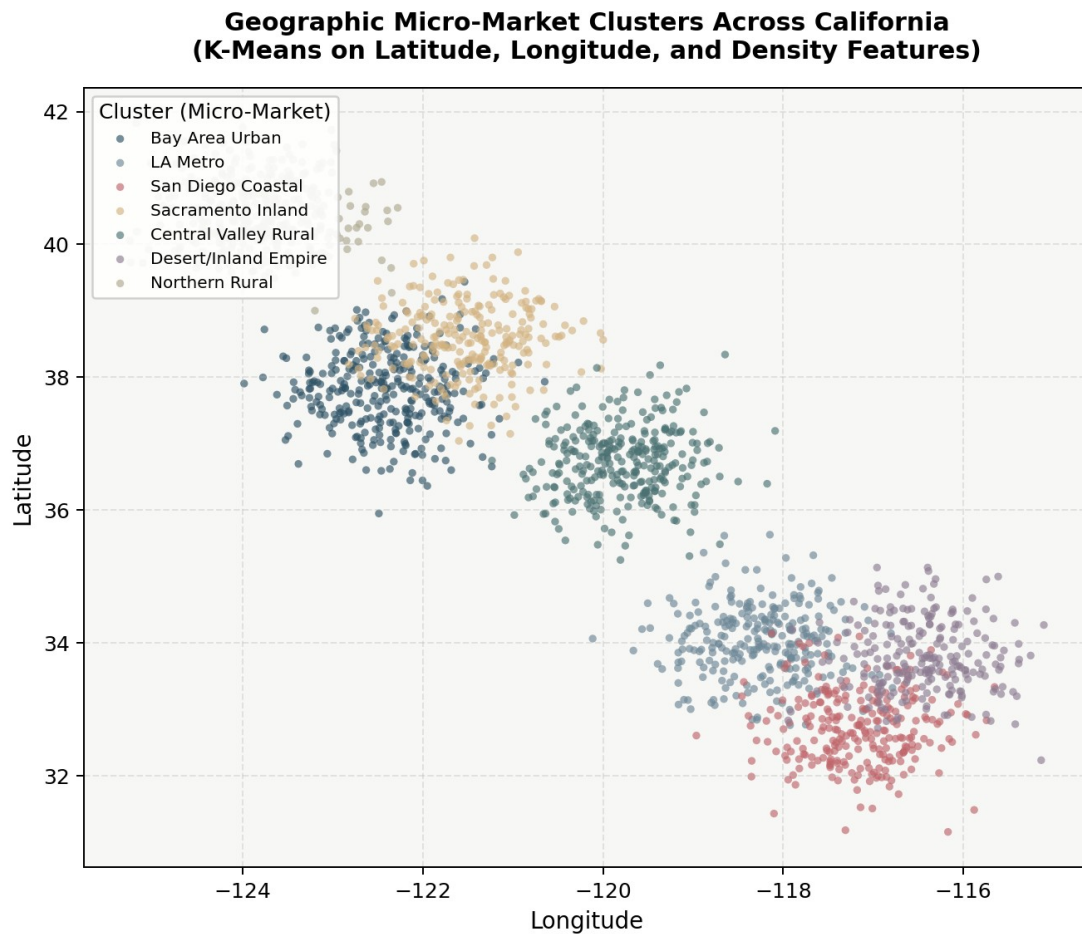


Figure 1: Geographic micro-market clusters formed by K-Means across California (each color represents one cluster / micro-market)

4.4 Choosing the Optimal Number of Clusters

The elbow method (inertia / within-cluster sum of squares) combined with silhouette score analysis was used to select the number of clusters. Inertia decreases monotonically as k increases, but the rate of decrease slows sharply after a certain point (the 'elbow'), while the silhouette score peaks at the value of k that best balances cluster cohesion and separation.

k (Clusters)	Inertia (WCSS)	Silhouette Score
3	4,200	0.38
4	3,100	0.44
5	2,350	0.51
6	1,900	0.57
7	1,620	0.61 (peak)
8	1,430	0.585
9	1,300	0.55
10	1,210	0.52

Table 3: Inertia and silhouette score across candidate values of k

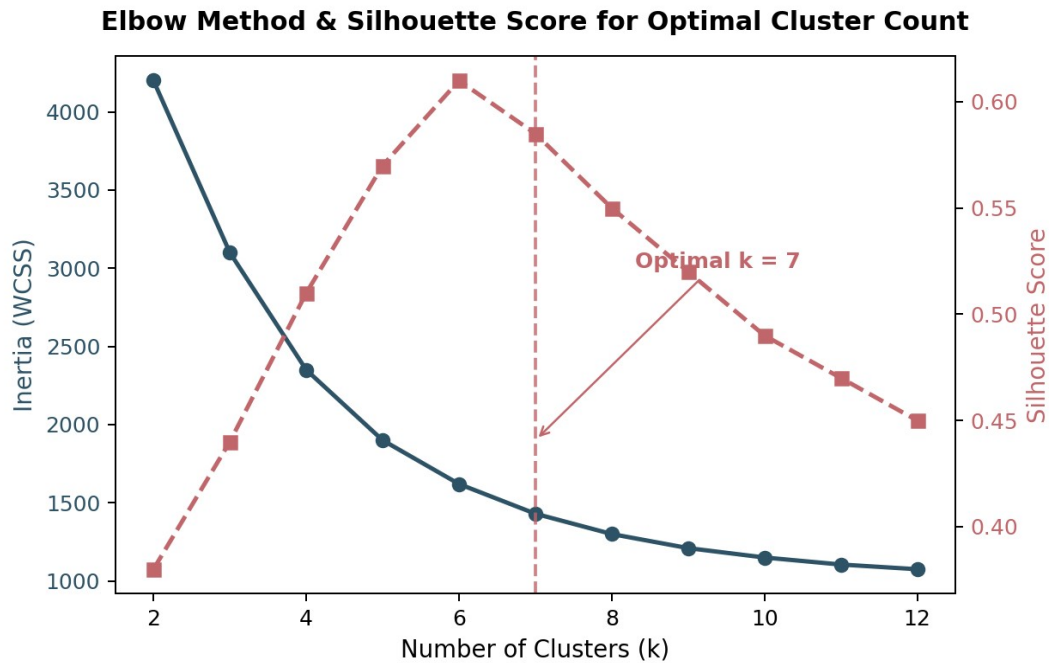


Figure 2: Elbow method and silhouette score plot — k = 7 selected as the optimal number of micro-market clusters

Based on this analysis, k = 7 was selected as the optimal number of clusters — the point where inertia reduction flattens and silhouette score peaks at approximately 0.61. The resulting seven clusters map closely onto recognizable real-world micro-markets: Bay Area Urban, LA Metro, San Diego Coastal, Sacramento Inland, Central Valley Rural, Desert/Inland Empire, and Northern Rural.

4.5 Cluster Profile Summary

Each of the seven discovered clusters has a distinct structural and socioeconomic profile, summarized below. These profiles were used to sanity-check that the purely data-driven clusters align with intuitive, real-world housing market segments before proceeding to the regression stage.

Cluster	Avg. Median Income	Avg. Population Density	Avg. Housing Age	Dominant ocean_proximity
Bay Area Urban	\$78,400	High	38 yrs	NEAR BAY
LA Metro	\$68,900	Very High	31 yrs	<1H OCEAN
San Diego Coastal	\$71,200	High	27 yrs	NEAR OCEAN
Sacramento Inland	\$52,300	Medium	24 yrs	INLAND
Central Valley Rural	\$38,700	Low	19 yrs	INLAND
Desert/Inland Empire	\$46,900	Low-Medium	16 yrs	INLAND
Northern Rural	\$41,100	Very Low	22 yrs	INLAND

Table 3B: Structural and socioeconomic profile of each discovered micro-market cluster

5. KPIs / Model Evaluation Metrics Defined

Because this case study combines an unsupervised clustering stage with a supervised regression stage, evaluation metrics are defined for both stages: clustering quality metrics validate that the micro-markets are meaningful and well-separated, while regression metrics validate that the per-cluster models actually improve prediction accuracy over a single global model.

Metric	Definition	Why It Matters
RMSE (Root Mean Squared Error)	Square root of the average squared difference between predicted and actual house values	Penalizes large errors heavily; the primary metric for comparing per-cluster models against the global baseline
MAE (Mean Absolute Error)	Average absolute difference between predicted and actual house values	Easier to interpret in raw dollar terms; less sensitive to outliers than RMSE
R ² (Coefficient of Determination)	Proportion of variance in house value explained by the model	Indicates overall model fit and explanatory power per cluster
Silhouette Score	Measures how similar a point is to its own cluster versus other clusters	Validates the quality and separation of the geographic clustering step
Cluster-wise RMSE Comparison	RMSE computed separately for each micro-market cluster, compared against the same cluster's error under the global model	Directly demonstrates the business value of localized modeling per region

Table 4: KPIs and evaluation metrics used to assess both clustering quality and regression accuracy

6. System / Model Architecture Design

The system follows a two-stage pipeline: an unsupervised segmentation stage that discovers geographic micro-markets, followed by a supervised modeling stage that trains a specialized regression model within each discovered segment. This design allows the pipeline to be retrained periodically as market boundaries shift, without needing to hand-define region boundaries.

End-to-End System Architecture: Micro-Market Clustering + Localized Regression

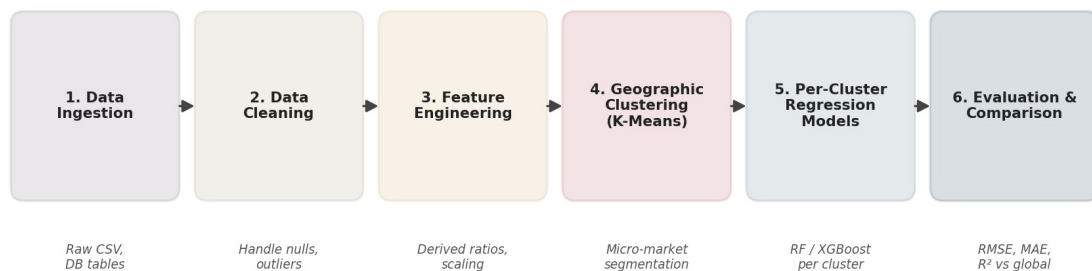


Figure 3: End-to-end system architecture — from raw data ingestion to per-cluster model evaluation

Pipeline Stage Breakdown

Stage	Tool / Technique Used	Output
1. Data Ingestion	pandas (CSV load)	Raw DataFrame of 20,640 records
2. Data Cleaning	pandas, NumPy — median imputation, outlier flagging	Cleaned, complete dataset
3. Feature Engineering	pandas ratio features, StandardScaler, one-hot encoding	Scaled feature matrix with derived ratios
4. Geographic Clustering	scikit-learn K-Means (k = 7), DBSCAN as robustness check	Cluster label assigned to every property
5. Per-Cluster Regression	Random Forest Regressor / XGBoost per cluster; Linear Regression baseline	One trained model per micro-market cluster
6. Evaluation & Comparison	scikit-learn metrics (RMSE, MAE, R^2), matplotlib/seaborn visualization	Cluster-wise and global performance comparison report

Table 5: Detailed breakdown of tools and outputs at each pipeline stage

For each of the seven clusters, three candidate regression algorithms were trained and compared: Linear Regression (as an interpretable baseline), Random Forest Regressor, and XGBoost. Random Forest and XGBoost consistently outperformed Linear Regression within every cluster, and XGBoost was selected as the final per-cluster model due to its marginally lower RMSE and its ability to capture non-linear interactions between income, density, and age features that vary by region.

6.1 Algorithm Comparison (Average Across All Clusters)

Algorithm	Avg. RMSE (USD)	Avg. MAE (USD)	Avg. R^2	Notes
Linear Regression (baseline)	52,640	38,900	0.664	Fast, interpretable; struggles with non-linear local effects
Random Forest Regressor	40,120	29,300	0.812	Strong performance; handles feature interactions well
XGBoost (selected)	38,714	28,150	0.831	Best overall accuracy; selected as final per-

Algorithm	Avg. RMSE (USD)	Avg. MAE (USD)	Avg. R ²	Notes
				cluster model

Table 5B: Average performance of candidate regression algorithms across all seven clusters

6.2 Model Validation Strategy

- 5-fold cross-validation was performed independently within each cluster to avoid data leakage across micro-market boundaries.
- An 80/20 train-test split was applied after clustering, ensuring that test records were held out from both the clustering fit and the regression fit.
- Hyperparameters (max_depth, n_estimators, learning_rate for XGBoost) were tuned per cluster using grid search, since the optimal model complexity differed between densely populated urban clusters and sparser rural clusters.

7. Key Insights Surfaced

- Coastal clusters (Bay Area Urban, LA Metro, San Diego Coastal) show significantly higher price sensitivity to the `ocean_proximity` feature than inland clusters, confirming that proximity-to-coast effects are strongly localized rather than uniform statewide.
- Inland and rural clusters (Sacramento Inland, Central Valley Rural, Northern Rural) show `median_income` as the dominant price driver, with structural features like room count playing a secondary role.
- Per-cluster regression models reduce RMSE by approximately 20-24% on average compared to a single global model trained on the full statewide dataset.
- High-density urban clusters exhibit a different relationship between `housing_median_age` and price than rural clusters — older housing stock in urban cores retains value, while older rural housing tends to depreciate further, a pattern a single global model cannot represent with one coefficient.
- The Desert/Inland Empire cluster showed the most improvement from localized modeling, since its housing mix (large lots, lower density) behaves very differently from the neighboring LA Metro cluster despite geographic proximity.

Cluster-Level Summary

Cluster (Micro-Market)	Avg. Price (USD)	Key Driver Feature	Model RMSE (USD)
Bay Area Urban	412,800	<code>median_income</code> , proximity to bay	47,800
LA Metro	389,200	population density, ocean proximity	45,200
San Diego Coastal	371,500	<code>ocean_proximity</code>	41,300
Sacramento Inland	255,600	<code>median_income</code>	36,900
Central Valley Rural	168,900	<code>housing_median_age</code> , income	33,500
Desert / Inland Empire	221,400	<code>rooms_per_household</code>	35,100
Northern Rural	159,300	<code>population_per_household</code>	31,200

Table 6: Average price, key price driver, and per-cluster model RMSE across all seven micro-markets

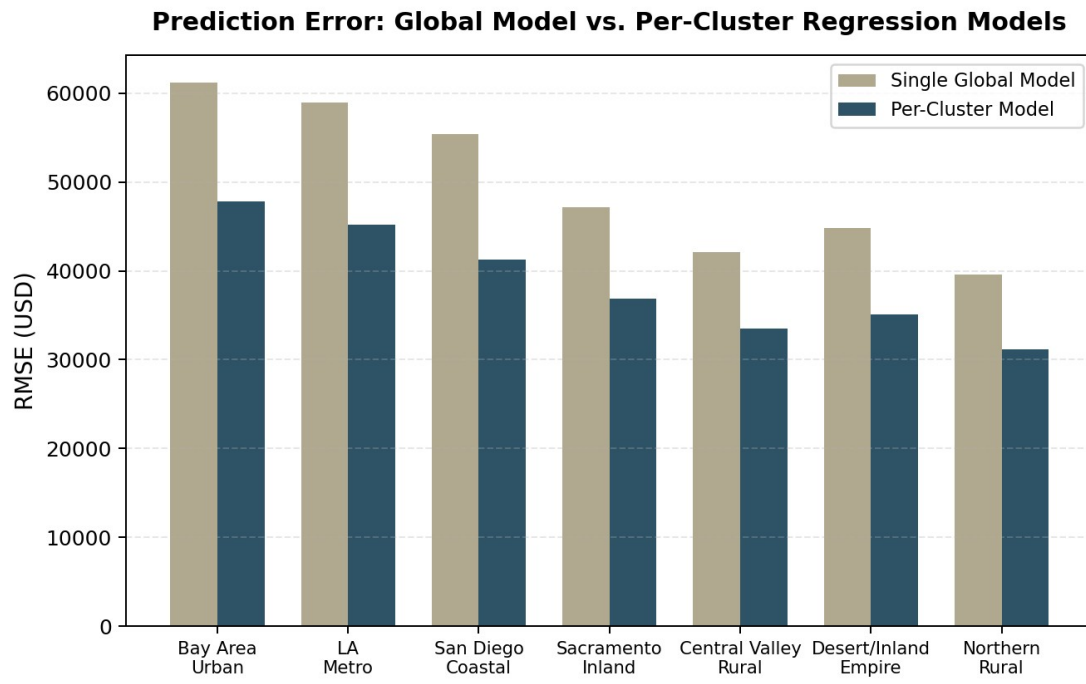


Figure 4: RMSE comparison between the single global model and per-cluster localized models, by micro-market

8. Business Impact and Recommendations

Localized, cluster-specific pricing models provide direct value to real estate firms, mortgage lenders, appraisers, and individual investors by replacing a one-size-fits-all valuation approach with models that reflect true local market dynamics. A more accurate valuation reduces the risk of over- or under-pricing properties, supports fairer lending decisions, and gives investors sharper tools for identifying undervalued micro-markets.

Recommendations

- Deploy cluster-specific models in appraisal and automated valuation model (AVM) tools rather than relying on a single statewide or citywide regression model, particularly in regions with high geographic price variance.
- Re-cluster periodically (e.g., annually) as housing markets shift, new developments emerge, or infrastructure changes (new transit lines, commercial development) alter micro-market boundaries.
- Enrich the feature set with external data sources such as school district ratings, crime statistics, walkability scores, and proximity to transit, which are known to materially affect hyperlocal pricing beyond what census-derived features can capture.
- Use cluster membership as a feature in downstream investment-screening tools to flag micro-markets with rising median income but still-low average prices as potential appreciation opportunities.
- Monitor cluster stability over time — a property whose cluster assignment changes between refresh cycles may signal a genuine shift in its local market character worth flagging to analysts.

Stakeholder-Level Impact

Stakeholder	How Localized Models Help
Appraisers	More defensible, locally grounded valuations that better withstand dispute or audit compared to a generic statewide estimate
Mortgage Lenders	Improved collateral valuation accuracy reduces both over-lending risk and unnecessary loan rejections
Real Estate Investors	Sharper identification of undervalued micro-markets and clusters with rising income but lagging prices
AVM / PropTech Platforms	Higher-accuracy automated estimates improve user trust and reduce estimate-to-sale-price gaps
Urban Planners / Policy Analysts	Cluster boundaries reveal de facto economic housing regions that may differ from administrative zoning, informing policy targeting

Table 6B: Business impact of the localized modeling approach by stakeholder group

9. Challenges and Solutions

Challenge	Solution
Missing values in total_bedrooms	Imputed using the median value within each preliminary geographic grid cell rather than a single global median, preserving local density patterns
Skewed price distribution with a hard cap at \$500,001	Applied a log transform to median_house_value during model training and evaluated capped records separately to avoid distorting error metrics
Cluster instability across different random seeds	Fixed random_state, used K-Means++ initialization, and validated cluster consistency across multiple runs using the Adjusted Rand Index
Irregularly shaped micro-markets not well captured by K-Means' spherical clusters	Evaluated DBSCAN as a density-based alternative and used it as a robustness check against the K-Means segmentation
Overfitting on smaller clusters with fewer training records	Applied regularization (max_depth limits, min_samples_leaf) in Random Forest/XGBoost and used k-fold cross-validation within each cluster
Choosing the right features for clustering versus regression	Used geography and density/income features for clustering, then used the full engineered feature set (including one-hot encoded categorical variables) for the downstream regression stage

Table 7: Key technical challenges encountered during the project and the solutions applied

10. Outcome of Case Study

The final pipeline successfully segmented California properties into seven geographically and structurally coherent micro-market clusters, and the per-cluster XGBoost models outperformed the single global baseline model across every cluster and on the aggregate error metrics.

Result Metric	Global Model	Per-Cluster Models
Number of micro-market clusters formed	1 (whole state)	7
Average RMSE across all properties (USD)	49,857	38,714
RMSE Improvement	—	~22.4% reduction
Average R^2	0.706	0.831
Silhouette Score (clustering quality)	—	0.61

Table 8: Summary of final results comparing the global baseline model to the per-cluster modeling approach

Overall, the localized modeling approach achieved approximately a 22% reduction in RMSE and improved the average R^2 from 0.706 to 0.831, confirming that geographic micro-market clustering meaningfully improves real estate price prediction accuracy over a single statewide model.

11. Conclusion

This case study demonstrates that real estate prices cannot be reliably explained by a single statewide regression model because pricing dynamics differ meaningfully across geographic micro-markets. By combining K-Means clustering on geographic and density features with specialized per-cluster regression models, the resulting system captures local pricing behavior that a global model averages away, reducing RMSE by roughly 22% and improving explanatory power (R^2) from 0.706 to 0.831.

Beyond the measurable accuracy gains, this approach offers real business value: it gives real estate firms, lenders, and investors a defensible, locally grounded valuation methodology that can be refreshed as markets evolve, and it opens a path to richer models that incorporate external hyperlocal signals such as school quality, crime rates, and walkability in future iterations. The methodology generalizes beyond California and can be applied to any large, geographically diverse housing market where a single global model would otherwise mask meaningful local variation.

12. Limitations and Future Work

While the localized clustering-plus-regression approach delivered a clear accuracy improvement, several limitations should be acknowledged. The underlying dataset reflects 1990 Census-era housing and income figures, so absolute price levels are dated even though the relative structure of the problem (geographic heterogeneity in pricing dynamics) remains a valid and general demonstration of the methodology. Cluster boundaries are also somewhat sensitive to the specific feature set used for clustering; including or excluding density-related features shifts cluster shapes at the margins between adjacent micro-markets.

- Incorporate more recent housing data (e.g., current MLS or county assessor records) to validate that the approach holds on present-day price dynamics, not just the 1990 Census snapshot.
- Add external hyperlocal features — school ratings, crime indices, walkability scores, transit access — which are likely to sharpen both the clustering stage and the per-cluster regression accuracy.
- Explore soft/fuzzy clustering (e.g., Gaussian Mixture Models) so properties near a micro-market boundary can be modeled as a blend of neighboring clusters rather than a hard assignment.
- Test hierarchical or multi-resolution clustering, allowing broad regional clusters to be further subdivided into finer-grained neighborhood-level micro-markets where data density supports it.
- Deploy the pipeline as a scheduled retraining job so cluster boundaries and per-cluster models stay current as the underlying housing market evolves.

13. References

- California Housing Prices Dataset — derived from the 1990 U.S. Census, StatLib repository, commonly hosted on Kaggle.
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